

SALAR SOHRABI

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SUMMARY

Senior Embedded Software Engineer with **15+ years** of experience delivering safety-critical automotive and embedded Linux platforms for production ECUs, infotainment and GNSS devices. Specialised in **AUTOSAR**, **ISO 26262**, **Linux BSP** and high-performance **C/C++17/20** across MCU and SoC platforms. Production software delivery for OEMs including **Volvo Cars**, **Volvo Trucks** and **Renault Group**. Strong hardware awareness including multilayer high-speed PCB design and board bring-up.

- **Embedded C/C++(17/20)**, Python, Rust, CUDA
- **Safety-Critical SW (ASIL-B/C/D)**
- **MCU: AURIX TC3xx**, NXP S32, Nvidia Orin
- **AOSP / Embedded Android**, BSP, HAL
- **AUTOSAR** (Classic/Adaptive), **ISO 26262**, ASPICE
- **Linux BSP**, Yocto, U-Boot, Kernel, Drivers
- **Robot Framework**, **HIL/SIL**, Jenkins, Docker
- **GNSS/RTK**, IoT, IPC, RTOS/**QNX** | License: B

EXPERIENCE

Sigma Technology (*Gothenburg, Sweden*) – *Senior Embedded Software Engineer* Dec 2022 – Present

- Designed high-performance **C++17/20** IPC framework using lock-free queues and zero-copy shared memory, reducing inter-process message latency on Nvidia Orin **Volvo Cars**
- Developed safety-critical software within **AUTOSAR**-based systems on Nvidia Orin/Xavier, DriveOS, **QNX**
- Implemented **IPC** and **UDP/TCP** communication components; **Robot Framework** integration testing on **HIL**
- Ensured **MISRA/Polyspace** static analysis compliance; built Python tooling for log analysis and validation
- Designed **CI/CD** pipelines (**Jenkins + Docker**); mentored junior engineers and performed code reviews
- Implemented **safety-related Embedded C** on **AURIX TC3xx** for powertrain ECUs **Volvo GTT**
- Designed SW architecture for safety-critical ECU subsystems per **ISO 26262 (ASIL-C/B)**; derived and documented safety requirements with full traceability
- Applied HW safety manuals (NXP/Infineon); implemented fault detection, monitoring, safe-state handling
- Developed peripheral drivers (GTM, ADC, MPU); configured **AUTOSAR** stack and **RTA-OS**
- Unit/integration testing with **Cantata** and Robot Framework; debugging with **Lauterbach** (T32)
- Developed **C/C++** components for truck simulation and multi-ECU integration; system-level validation and debugging within Linux-based environments **Volvo Trucks**
- Designed **HEBAT** battery monitoring firmware (**STM32**, custom binary protocol with CRC-8) and .NET/ASP.NET backend with Blazor Server and **SignalR** real-time streaming **In-house R&D**
- Developed battery health scoring algorithms (voltage, stability, temperature, current) and scenario-based load simulation with programmable switching schemes
- Full telemetry pipeline (USB → parser → backend → SignalR → browser); **HIL**-validated, safety-enforced

Capgemini Engineering (*Fundão, Portugal*) – *Senior Embedded Software Engineer* Dec 2020 – Dec 2022

- Performed **end-to-end Android platform bring-up** on embedded hardware (**AOSP**-based infotainment)
- Developed **BSP**, **kernel modules** and **HAL** layers; built **HIDL/AIDL** binder interfaces in **C++**
- Developed **SELinux** policies for Android services; customised Android build system (Soong, device trees)
- **Embedded Linux** user-space daemons: multi-threading, Sockets, IPC mechanisms, RPCs, synchronisation
- Developed automotive software using **C/C++**, **AUTOSAR**, **ISO 26262 (ASIL-B/C)** **Renault Group**
- Safety requirements analysis and **ASIL decomposition**, FMEA contribution; **GoogleTest**, **MISRA/Polyspace**

Silix co. – *Startup Co-founder / Embedded Linux & BSP Lead* Feb 2016 – Dec 2020

- Built 500+ channel **GNSS/RTK OEM** module; led 3 engineers from architecture to mass production in one year
- Developed **Embedded Linux BSP** from scratch: **U-Boot**, kernel, device tree overlays, rootfs on Cortex-A53
- Wrote **Linux drivers**: Bluetooth, high-speed USB, 4G WWAN, GPS, serial; boot time <10 s via **Yocto**

- Hand-held GPS (<2 cm accuracy) using uBlox **RTK** with IMU dead-reckoning for GNSS-denied areas
- Designed multilayer high-speed PCBs (6+ layers, controlled impedance) in Altium; full HW/SW co-design

Earlier Roles – *Embedded Hardware/Software Engineer*

Jun 2008 – Feb 2016

- **Medaria** (2014–2016): Led HW/SW for CRYO/3D-Fractional medical devices (**IEC 60601**); Cortex-M4, 600 V PWM; multilayer PCB design
- **Boreshsazan** (2012–2014): DLP/SLA 3D-Printer firmware (FPGA/MCU, sub-0.1 mm accuracy); 3-axis CNC controller; PCB design with signal integrity alongside noisy power stages
- **Noavaran Moj** (2008–2012): 500 W offline PSU with PFC (>90% eff., PF=0.94); digital DC electronic load (CC/CV/CR/CP); battery chargers/analysers via EIS; led team to mass production

SKILLS

Languages	C, C++(14,17,20), Python, Rust, CUDA, C#/.NET, Assembly MISRA/Polyspace, Robot Framework, JS (Node.js, TS), HTML/CSS
Automotive Safety	Classic/Adaptive AUTOSAR, MCAL, ARXML, ISO 26262 (ASIL-B/C/D), ASPICE Functional Safety Engineering, Safety Requirements Derivation, ASIL Decomposition Safety Analysis (FMEA/FTA), Fault Detection, Safe-State Handling, CANoe, SOME/IP
Architecture	Software Architecture, HW/SW Co-design, Distributed Systems, Real-Time Event Processing
Linux	Kernel, Rootfs (Yocto/Buildroot), U-Boot, Device Tree, Shell, SELinux
Android	AOSP, Platform Bring-up, BSP, HAL (HIDL/AIDL), Binders, SELinux
RTOS	QNX, Linux (PREEMPT_RT), RTA-OS, FreeRTOS, uCOS
Tools	Cantata, Lauterbach, GDB, CMake, Bazel, UML, Design Patterns
DevOps	Jenkins, Docker, GitLab CI, containerised dev environments, Grafana
Version	Git, Gerrit, GitLab, SVN V-model, TDD, Unit Testing, Mocks
MCU/SoC	Nvidia Orin/Xavier, AURIX TC3xx, NXP S32, STM32MP15x, i.MX6/8, AM335x STM32F7/H7, LPC43x, ZYNQ (XC7Z0xx), Artix-7/Virtex/Spartan, Cyclone, TMS320
Hardware Protocols	Altium Designer, Multilayer PCB (controlled impedance), Board Bring-up, Signal Integrity CAN, CANopen, LIN, UDS, SOME/IP, MODBUS, RS-485, Ethernet, PCIe, MQTT 4G/LTE, LoRA, ZigBEE, WLAN, BLE, SDR (HackRF/USRP)
Spoken	Fluent: English Basic: Swedish Driver License: B

PROJECTS & RESEARCH

Price Structure Engine (C++20, Python, PyTorch). Real-time event processing framework for market structure analysis. Lock-free data structures, Python bindings, ML pipelines for financial time-series.

Gesture Recognition for HMI (C++, OpenCV, OpenCL, MATLAB). Industrial machine control via stereoscopy and IR camera; 10 gesture classes on PC, SBC and ZYNQ hardware.

Smart Home IoT Controller (C++, Linux, Python, React Native). Star-topology IoT on Cortex-A53/A72; LoRA, ZigBEE, WLAN, MQTT. Led 5 engineers to mass production; 10+ installations; boot <10 s via Yocto.

Satellite Receiver/Tracker (ZYNQ, Cortex-H7, Altium). Schematic, multilayer PCB and firmware for 3-axis servo tracker. 0.1° accuracy, 0.05°/s slew rate on moving-base platform.

EDUCATION

M.Sc. Electronics Engineering , Sharif University of Technology	2005 – 2008
B.Sc. Electronics Engineering , Iran University of Science and Technology	2001 – 2005

ADDITIONAL

Ranked 17th in national M.Sc. entrance exam (14 000 candidates) | Top 0.5% in National University Entrance Exam (200 000 candidates) | **Interests:** High-performance computing, embedded AI, algorithmic systems